AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM358
Machine Learning &Artificial intelligence

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 9 /April/ 2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 120 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

AOU EXAM PAPER

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Part -I: Question 1: Choose the best correct answer of all the following: (15 Marks)

1 ..... knowledge deals with how a question is answered in AI or how a process is performed.

A Declarative
B Instructional
C All of the above
D Procedural

2 ..... tend to provide more information, but at the expense of increased power consumption and with a danger of interference when multiple sensors are used at the same time.

A Active sensors
B mobile sensors
C Passive sensors
D rigid sensors

3 ..... can be defined as "The ability of some unit to alter its longer-term behavior patterns in response to challenges from its environment"

A interaction
B adaptation
C emergence
D selection

4 ..... Means "Is the algorithm guaranteed to find a solution when there is one?"

A Time complexity
B Space complexity
C Completeness
D Optimality

5 .....means "How long does it take to find a solution?"

A Optimality
B Space complexity
C Time complexity
D Completeness

6 ..... can be defined as "Given a set of goals, construct a sequence of actions that achieves those goals "

A Acting
B Planning
C All of the above
D Performing

7 ..... just like DFS, except never go deeper than some depth d

A Best first
B Depth-limited
C BFS
D A*

8 in the ....., nodes are expanded in the same order in which they are generated

A Best first
B BFS
C A*
D DFS

9 ..... can be defined as "is the collective behavior of decentralized, self-organized systems, natural or artificial"

A adaptation
B selection
C swarm intelligence
D interaction

10 ..... involve building a possible solution step by step, at each step picking the most possible candidate from the set of possibilities

A swarm intelligence
B Local search algorithms
C Construction algorithms
D adaptation

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Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

A.

At a given point of optimization time and given an ant from ACO swarm, the ant found 4 possible routes at its journey. The pheromone amounts on each route are :

9.0 3.0 8.0 5.0

If the length of each route which is inversely proportional to the route desirability is known to the ant and lengths are :

54.0 55.0 66.0 8.0

Calculate the probability of selecting each route by that given ant
Assume Alpha=0.9 and beta= 0.8

B. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s, t in order for a given tree as:

s	A	A	A	B	B	B	C	C	C	C	D	D	E	F	F	F	G	G	H	H	H	I	I	I	J	J	J	K	K	L	L	L	L	M
t	B	C	D	E	F	G	H	I	J	K	L	M	37	25	16	32	31	36	39	5	41	2	18	10	19	48	21	33	28	46	43	27	26	7	

Given the aforementioned tree, use minimax algorithm to evaluate each node in the tree.

Question 3: (15 marks)

A. Generate the search tree for *the stone game(4,2)* and use minimax algorithm to calculate the value of each node hence decide the winner for that game assuming rational steps.

B. Having the following XO sequence of states a long with their values

X O O - O X - - X	0.500
X O O - O X - X X	0.526
X O O O O X - X X	0.510
X O O O O X X X X	1.000

Assume a learning rate of 0.51 what will be updated values adopting gradient-based state value update with each move.

Question 4: (15 marks)

a. Explain how temporal difference (TD) leverages incremental state value propagation and simple gradient updates in temporal difference learning.

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b. A tree can be defined by two vectors. The first vector is the parents vector -s. The second vector is the children vector -t. Each entry in t matrix is a child to the corresponding parent in the same index in the s matrix. Assuming s and t in order for a given tree as:

s	A	A	A	B	C	C	C	D	F	F	H	H	I	I	I	I
t	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q

Given the aforementioned tree, apply DFS to traverse the tree showing all agenda content and traced paths until reaching the goal state if exist assuming that the goal state is I

End of questions

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